Detection-Theoretic Decision Rules for AI-Agent Monitors

Overall summary: what we tried and what we found

October 2026

In one paragraph. AI-control monitoring has invested heavily in better *monitors* (stronger LLM judges, better prompts) but almost nothing in the *decision layer*: how per-step scores become audit decisions. Everyone takes the max over steps and compares it with one global honest-run quantile. Starting from detection theory (Neyman–Pearson, CFAR, CUSUM/SPRT), we reviewed 92 papers and 101 lab and blog sources, ranked 10 project ideas, and ran the top three on public control datasets and new monitor scoring ($195.56 total OpenRouter spend). The results:

1. **Accumulation vs max.** Textbook sequential tests do *not* beat max at a matched false-alarm budget on real score streams. Accumulation wins clearly in one regime: *cross-session* evidence from *history-free* per-session monitors (+0.15 TPR). Which rule wins is predicted by measurable stream properties.
2. **Per-context thresholds.** The obvious radar import, CFAR, is the *worst* rule against an attacker who chooses its context. Damage-equalizing (Stackelberg) thresholds are best. Robust order-statistic baselines estimated from the other contexts resist poisoning.
3. **Detector-aware attackers.** An LLM attacker told the exact detector did not evade better against a weak monitor. It condensed its attacks and got caught more. Testing the covert-adversary theory needs a strong monitor.

1 What we did

1. **Background research.** We read 92 arXiv papers in full (2009–Oct 2026) and 101 lab and community sources: Redwood, Apollo, UK AISI, Anthropic, OpenAI, GDM, METR, LessWrong/AF, and security vendors. We red-teamed the starting idea of building CFAR- and CUSUM-style monitors (`BACKGROUND_SUMMARY.md`).
2. **Idea generation.** We ranked 10 ideas by novelty, impact and feasibility (`IDEAS.md`) and ran the top three:
   - matched-budget decision rules;
   - per-context thresholds;
   - monitor-aware attackers.
3. **Infrastructure.** We built a private repository (`gregkocher/agent-monitor-detection`) with:
   - loaders for six public datasets;
   - a per-step monitor scoring harness (OpenRouter, logprob-expected scores, resumable cache);
   - a detector library (12+ decision statistics with matched-FPR calibration);
   - a spend ledger enforcing a $500 cap.

   Everything ran on the laptop.

2 The landscape (from the literature review)

- **Uniform practice.** Research evaluations (BashArena, LinuxArena, MonitoringBench, Ctrl-Z, SHADE-Arena, Async Control, and others) set a 98th–99.9th-percentile honest threshold and aggregate steps by max. Deployed systems use fixed severity cutoffs: Watcher (severity ≥ 6), Claude Code auto mode (2-stage classifier plus a "3 consecutive / 20 total denials" rule), OpenAI's async severity tiers. Nobody reports ARL, detection delay, or anytime-valid guarantees.
- **Adaptivity sits in the protocol, not the threshold.** Examples: decaying audit budgets (Ctrl-Z), Bayesian credence across tasks (Wen et al.), resampling, cascades.
- **Closest prior art, and its limits:**
  - a June 2026 preprint claiming CUSUM/SPRT on control score streams: synthetic streams, and its code does not implement sequential tests;
  - CUSUM for benign agent failures (CURA, Sentinel) and for jailbreaks (SafeDream, CPD);
  - anytime-valid tests for task correctness (e-valuator).

  No prior work applies these to real control-setting suspicion streams at matched budgets, uses per-context nulls, or tests against a detector-aware attacker.
- **Warnings from the literature:**
  - k-strikes accumulation *failed* three times;
  - thresholds do not transfer across contexts or model families (0–94% realized audit spend);
  - attack selection costs 20–28 pp of safety;
  - theory gives a covert-attacker bound: damage $\sim \sqrt{\text{ARL}}$.

3 Results at a glance

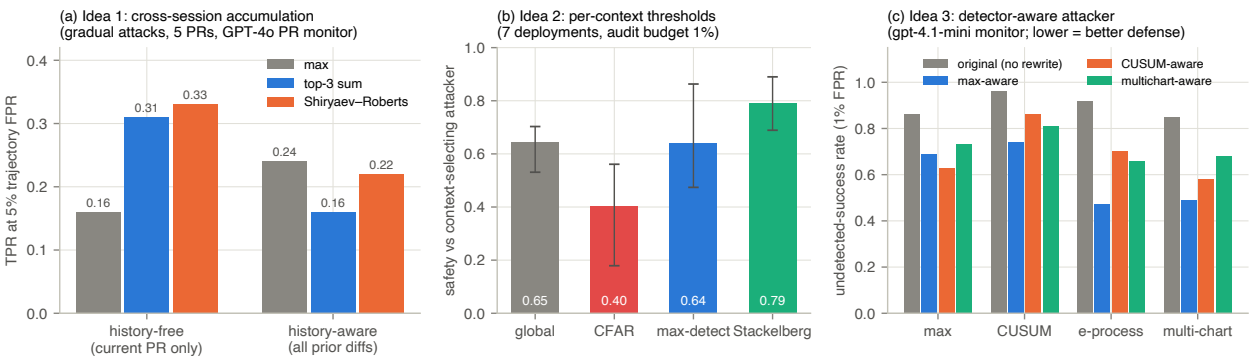


Figure 1: One key result per idea. (a) Cross-session accumulation beats max only with a history-free per-PR monitor. (b) Against an attacker that picks its context, CFAR is worst and Stackelberg (damage-equalizing) best. (c) Detector-aware rewriting *lowered* undetected success compared with the original attacks.

4 Idea 1: matched-budget decision rules

Question. At equal trajectory FPR, do CUSUM, Shiryaev–Roberts, GLR, e-processes or top-k rules beat max? Pre-registered hypothesis: accumulation helps only for history-free ("innovation")

monitors, because history-aware scores already integrate evidence ("double counting").

Approach.

- 46 existing stream sets: aksh-n Gemini streaming judges, the 12-monitor LinuxArena sweep, MRT, and persistent-state per-PR monitors.
- A new controlled experiment: 775 MonitoringBench attacks and 550 honest runs scored per step by the same `gpt-4.1-mini` monitor with full history vs none.
- Honest-only calibration with cross-fitting and randomized thresholds; cluster-bootstrap CIs.

Findings.

- **Max-type rules win in most sets.** Max, 2nd-max or length-aware max matched or beat every accumulator in 86% of aksh-n, 67% of LinuxArena and 83% of MRT sets at 1% FPR. Accumulation gain falls with honest autocorrelation (Spearman -0.57).
- **Cross-session accumulation works with a history-free monitor.** With the history-free per-PR monitor, top-3 sum / Shiryaev–Roberts beat max by $+0.15$ TPR on gradual 5-PR attacks. The gain disappears with the history-aware PR monitor.
- **The controlled MonitoringBench experiment did not confirm the simple hypothesis.** `gpt-4.1-mini`'s history-aware honest scores were not cumulative; instead, history made it *re-flag attacks*. Best rules: top-3 sum / 2nd-max (≈ 0.19 TPR at 1% FPR vs max 0.17).
- **Refined mechanism.** Across the 46 sets, the gain from accumulation tracks:
  - attack repeat-spike informativeness ($+0.47$);
  - honest spike persistence (-0.67).

  Both are measurable before choosing a rule.
- **CUSUM's k is critical:** $k = 0.5$ fails; $k \approx 1.5$–2 matches max. A length-aware max is a large free win when lengths differ (MRT $0.21 \rightarrow 0.58$).

5 Idea 2: per-context thresholds

Question. At a fixed global audit budget, should thresholds be set per context (CFAR, the radar idea), globally, or by a damage-equalizing Stackelberg allocation? And what about poisoning of per-context baselines?

Approach.

- Real panels: Gemini judges over 7 settings and 12 ControlArena setting×agent contexts; LinuxArena environments, task types and agent models; the 12-monitor sweep; MRT tasks; MonitoringBench clusters.
- Budget-matched rules; uniform, informed context-selecting, cell-selecting and damage-weighted attackers.
- Small-sample subsampling; one-shot and boiling-frog poisoning.

Findings.

- **Heterogeneity is real across deployments, small within one.** A global 5% threshold yields 0–9% per-setting FPR ($\tau = 1.34$, $p < 0.001$) across deployments but 2–7% within LinuxArena. Agent model is irrelevant. Environment effects are shared by all 12 monitors (Spearman 0.70), so they come from the trajectories.
- **Against an attacker that picks contexts:**

  - CFAR is **worst**: safety 0.40 vs global 0.64 at a 1% budget (ControlArena 0.22 vs 0.60);
  - **Stackelberg is best**: 0.79 vs 0.64 at 1%, 0.97 vs 0.89 at 5%, at ≤ 0.02 uniform-TPR cost.
- **Small samples break per-context quantiles** ($7\times$ over-spend at $n = 10$). Partial pooling with cross-validated strength stays near budget.
- **Poisoning.** CFAR recalibration breaks at a poisoned share $\varepsilon \approx B$ (TPR $0.95 \to 0.37$ at 2%). OS-CFAR with a leave-one-context-out tail holds 0.77–0.86 and still tracks genuine drift. Outlier filters only slow a boiling frog.

6 Idea 3: monitor-aware attackers

Question. Can an attacker that knows the detector (max, CUSUM(k, h), e-process, multi-chart) restructure real multi-step attacks to evade it without breaking them? Does undetected damage scale like $\sqrt{\text{ARL}}$, as covert-adversary theory predicts?

Approach.

- 150 MonitoringBench S2–S4 attacks rewritten by `gpt-4.1`, which was told the monitor prompt and the detector rule and given up to two rounds of feedback.
- Validity checked by LLM-judge checklists and literal-target checks.
- Full attacker $\times$ detector cross at matched FPR, with a `gpt-4.1-mini` monitor.

Findings.

- **Aware rewriting did not help the attacker; it hurt.** At 1% FPR, undetected success against max was 0.86 for the originals vs 0.63–0.73 for the aware arms. The attacker condensed attacks (length ratio 0.58–0.83), raising peak suspicion from 5.8 to 7.6.
- **Detector ranking.** CUSUM alone is the most evadable detector. Max and the multi-chart OR are strongest; the OR closes CUSUM's gap but ties max.
- **Adaptive feedback** helps the attacker (+0.10 to +0.17 over one-shot), but not past the originals.
- $\sqrt{\text{ARL}}$ **scaling** is not observed (slope ≈ 0). This weak monitor leaves attacks below the detectability band at every ARL.
- **Validity is high:** 98–99% judge preservation. No Docker replay (disk and VPN constraints).

7 Cross-cutting lessons for practitioners

1. **Default trajectory rule:** max with a length-aware null, plus 2nd-max or top-3 as a cheap upgrade. Do not deploy CUSUM alone. If combining, calibrate an OR multi-chart *jointly*.
2. **Accumulate across sessions with history-free per-session judgments.** This is the one clean gain, and it addresses the cross-trajectory "diffuse/distributed" concern that labs name as open.
3. **Choose the rule from data:** estimate honest spike persistence and attack repeat-spike informativeness first.
4. **Per-context thresholds:** measure heterogeneity (τ) first. If contexts differ and attackers can pick, use Stackelberg (damage-equalizing), not CFAR. With few honest runs, use partial pooling. If the agent can touch calibration data, use OS-CFAR with a leave-one-context-out tail, or global.

5. **Evaluate against detector-aware attackers.** But note that a weak monitor makes such tests uninformative, because dilution already wins.

8 What is new here (to our knowledge, per the literature review)

- The first matched-budget comparison of sequential and anytime-valid statistics vs max on real AI-control score streams (46 sets plus a controlled history experiment). It answers Monitoring-Bench's open question.
- Matched-budget evidence that cross-session accumulation of history-free judgments beats max. The persistent-state source paper tested only a failing k-strikes rule.
- A measurable mechanism (honest persistence vs attack repeat-spike informativeness) explaining when accumulation helps, including the earlier negative results.
- The first per-context threshold study for control monitors, showing CFAR's failure under attack selection and the Stackelberg alternative, with poisoning-robust baselines.
- The first evaluation of sequential detectors against detector-aware LLM attackers, with a negative result about condensation.

9 Budget, compute and artifacts

Item	OpenRouter spend	Notes
Idea 1	$109.56	per-step scoring of 775 attacks + 550 honest, two history modes; nano arm
Idea 2	$17.11	trajectory-mode scoring of MonitoringBench
Idea 3	$68.36	refiner, judge, checklists, scoring of rewritten attacks
Development	$0.52	harness validation
Total	**$195.56**	of the $500 cap (MATS account)

- **Compute.** Everything ran on the laptop. No RunPod time was used beyond a test pod deleted within minutes.
- **Artifacts**, all in `~/Desktop/monitoring-research/`:
  - `BACKGROUND_SUMMARY.md`, `IDEAS.md`, `background/papers/` (reading cards);
  - `agent-monitor-detection/` (code, `results/t{1,2,3}/`, `EXPERIMENT_LOG.md`);
  - `reports/` (these PDFs).

10 Limitations and recommended next round

- **Weak monitors.** The re-scored monitors are cheap (`gpt-4.1-mini/nano`); strong-monitor streams came only from existing datasets.
- **Narrow positive result.** The cross-session result rests on one monitor (gpt-4o) and 5-PR runs.
- **Attack validity is not verified by execution.** Idea 3's attacks were checked by LLM judge, not replay.

Recommended second round (about $300 of budget remains):

1. Confirm the cross-session result with a second monitor family and longer PR sequences ($10–40).
2. Re-run Idea 3 with a strong monitor and an anti-condensation attacker instruction. This is the only way to test the covert-adversary regime and accumulation vs pacing ($100–180).
3. Replay about 50 rewritten attacks in BashArena Docker sandboxes on a pod, once SSH is reachable.

Detailed reports (same folder):

- `01_Idea1_Matched_Budget.pdf`
- `02_Idea2_Per_Context_Thresholds.pdf`
- `03_Idea3_Aware_Attacker.pdf`